CHEMICAL CHANGE

ISOTOPES

Isotopes: atoms of one element with the same number of protons but different numbers of neutrons.

The number of protons is fixed for any given element, however the number of neutrons can often vary. The Mass Number of an element is always a whole number but because there can be more than one type of the same element, the RAM of an element (this is the number underneath the symbols on the periodic table) is often not a whole number because it is an average of two or three isotopes of the same element.

Example

Chlorine has two isotopes and can have either 18 neutrons or 20 neutrons (the number of protons 'Z' is always 17).

$^{35}_{17}Cl$ or $^{37}_{17}Cl$ (*Chlorine Isotopes*)

In nature any random sample will contain 3/4 (75%) of $^{35}_{17}Cl$ and 1/4 (25%) of $^{37}_{17}Cl$.

To calculate its Relative Atomic Mass (RAM) we do the following.

$$\begin{array}{rcl} & & 75\% \times 35 \\ & + & \underline{25\% \times 37} \\ \text{RAM} & = & 35.5 \end{array}$$

On the periodic table you can see that the RAM is 35.5

Try this:

Natural Silicon in silicon ore contains 92% $^{28}_{14}Si$, 5% $^{29}_{14}Si$ and 3% $^{30}_{14}Si$. Calculate its the RAM of silicon in nature.